

ER DIAGRAM

ER Diagram for shopez ecommerce:

- Users entity
- Products entity
- Cart entity
- Orders entity
- Relationships between the entities (1:N and N:1)
- A brief list of key attributes for each entity

If you need a professional ER diagram with crow's foot notation (like diagrams created in MySQL Workbench, Draw.io, or Lucid chart) for inclusion in a project report or thesis, I can generate a polished version as well.

CODE:

```
from reportlab.platypus import SimpleDocTemplate, Paragraph

from reportlab.lib.styles import getSampleStyleSheet

from reportlab.lib import colors

from reportlab.graphics.shapes import Drawing, Rect, Line, String


# Create PDF

doc = SimpleDocTemplate("ER_Diagram_Ecommerce.pdf")

styles = getSampleStyleSheet()

elements = []


# Title

elements.append(Paragraph("<b><font                size=16>E-Commerce                ER  
Diagram</font></b>", styles["Title"]))
```

```
# Drawing Area
```

```
drawing = Drawing(550, 350)
```

```
# Entity Boxes
```

```
entities = {
```

```
    "Users": (30, 260),
```

```
    "Cart": (210, 260),
```

```
    "Orders": (390, 260),
```

```
    "Products": (210, 70)
```

```
}
```

```
# Draw Entities
```

```
for entity, (x, y) in entities.items():
```

```
    drawing.add(Rect(x, y, 120, 50, strokeColor=colors.black, fillColor=None))
```

```
    drawing.add(String(x + 30, y + 22, entity, fontSize=12))
```

```
# Relationships
```

```
# Users → Cart
```

```
drawing.add(Line(150, 285, 210, 285))
```

```
drawing.add(String(170, 292, "1 : N", fontSize=9))
```

```
# Cart → Orders
```

```
drawing.add(Line(330, 285, 390, 285))
```

```
drawing.add(String(345, 292, "1 : N", fontSize=9))
```

```
# Cart → Products
```

```
drawing.add(Line(270, 260, 270, 120))
```

```
drawing.add(String(278, 190, "N : 1", fontSize=9))
```

```
# Orders → Products
```

```
drawing.add(Line(450, 260, 330, 120))
```

```
drawing.add(String(390, 190, "N : 1", fontSize=9))
```

```
elements.append(drawing)
```

```
# Entity Attributes
```

```
text = ""
```

```
<b>Entities and Attributes</b><br><br>
```

```
<b>Users</b><br>
```

```
• UserID (PK)<br>
```

```
• Name<br>
```

```
• Email<br>
```

```
• Password<br>
```

```
• Role<br><br>
```

```
<b>Products</b><br>
```

```
• ProductID (PK)<br>
```

```
• Name<br>
```

```
• Category<br>
```

```
• Price<br>
```

```
• Stock<br><br>
```

```
<b>Cart</b><br>
```

```
• CartID (PK)<br>
```

- UserID (FK)

- ProductID (FK)

- Quantity

Orders

- OrderID (PK)

- UserID (FK)

- ProductID (FK)

- TotalAmount

- OrderDate

- Status

""""

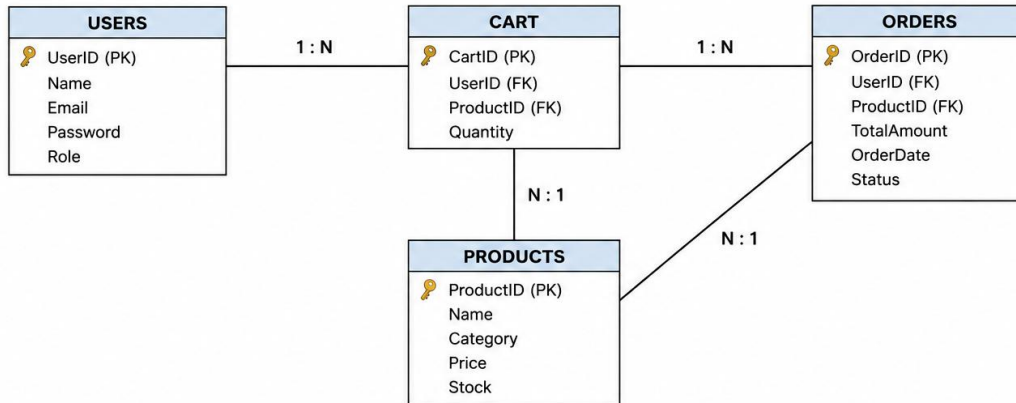
elements.append(Paragraph(text, styles["BodyText"]))

Build PDF

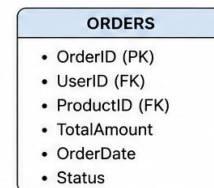
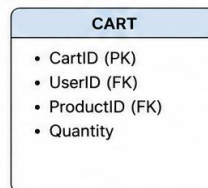
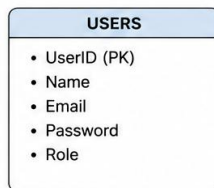
doc.build(elements)

print("ER_Diagram_Ecommerce.pdf generated successfully.")

E-COMMERCE ER DIAGRAM



ENTITIES AND ATTRIBUTES



PK – Primary Key
FK – Foreign Key

1 : N One to Many
N : 1 Many to One

Primary Key
Relationship